



## Cemo CPH 505

**Typical applications:** Deep mine dewatering and public utility water transfer

CEMO Multistage Centrifugal Pumps are used for large volume water transfer in both deep mining dewatering applications including water transfer for public water utilities. Advantages include high mechanical and volumetric efficiencies, large flow rates and high heads. Materials of construction are sand and die cast PB1 Phosphor bronze (impellers, diffusers, shaft sleeves, neck rings, neck bushes, balance and counter disc components. All stage casings, suction and delivery covers are cast from A-2 Steel or Gr17 Cast Iron or SG42 nodular iron as per client's requirements or specifications. Anti-abrasive coatings such as HVOF tungsten carbide, ceramic coatings and chrome plated coatings are used on wear susceptible wetted components or rotating element assemblies. All rotating element individual components and assemblies are balanced to ISO 1940/1 prior to assembly. All pump shafts are non-destructive tested and heat treated to relieve residual stresses.

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Tel: +27 618-2003 E-Mail: [sales@cemo.co.za](mailto:sales@cemo.co.za)

Physical Address: Crown Building, 64 Hanau Street, Jeppestown, JHB

Standard Trading Terms, Conditions and Warranties Apply And Are Available On request



All pumps are independently tested at The South African Bureau of Standards prior to delivery and a detailed Data Pack of material certification, non-destructive tests, balancing certification and verification of pump performance is documented and supplied on pump delivery to client.

All pumps and spares fully manufactured (machining and castings) at our Johannesburg SA works to ISO 9001:2015 requirements

### **Typical materials of construction**

#### **Suction cover:**

A2 Steel with right hand flange orientation

#### **Delivery cover:**

A2 Steel with vertical discharge flange configuration

Note: All flanges (rated to 15 Mpa) are to ANSI spec or drilled as per client's requirements

#### **Stage casings:**

Cast and stress relieved A2 Steel

#### **Balance Valve Assembly:**

PB1 Bronze / 431S29 Forged balance disc with Cast Iron Counter Disc fitted with PB1 Bronze Wear Sleeve and Stainless Steel Throttling Bush

#### **Wear Faces:**

PB1 Bronze / Stainless Steel or hard faced with HVOF Tungsten or Stellite Welded as per client's requirements

#### **Binding bolts & nuts:**

Certified 080-M40 (BS970-1991) steel threaded both ends

#### **Shaft:**

Forged 817-M40 or 709-M40 Condition "T" ultrasonically non-destructive tested and stress relieved for maximum tensile strength

#### **Impellers / Diffusers / Shaft Sleeves:**

PB1 bronze, chill cast (impellers & diffusers) or 316 Stainless Steel or PB1 bronze as per client's Shaft sleeve requirements. If requested abrasive resistant

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HVOF Tungsten coating on all wear areas and tips including Shaft sleeve wear areas (all Tungsten coatings minimum 200 Microns per side)

Neck Rings:

PB2, chill cast and Tungsten coated to suit client's requirements

Bearing Housings:

Nodular Iron SG42 with water cooling jacket

Non Drive End Bearing Bracket:

Nodular Iron SG42 with water cooling jacket

Bearings:

Cast Iron housings lined with WM90 white metal. Bearing housings fitted with water cooling jackets for cooler bearing operation.

Optional Equipment:

- Digital balance water return flow meter
- Pump monitoring kit (High and low pressure gauges including high pressure monitoring cock)
- Coupling – Grid or Gear type as per client's requirements
- Pump and motor baseplate assembly
- Wear Limit Indicator Switch – mechanical or proximity type to identify Wear Face deterioration
- Electrically actuated pump isolation valves and pump Non-Return valves

Pump protection: (compulsory supply with all pumps)

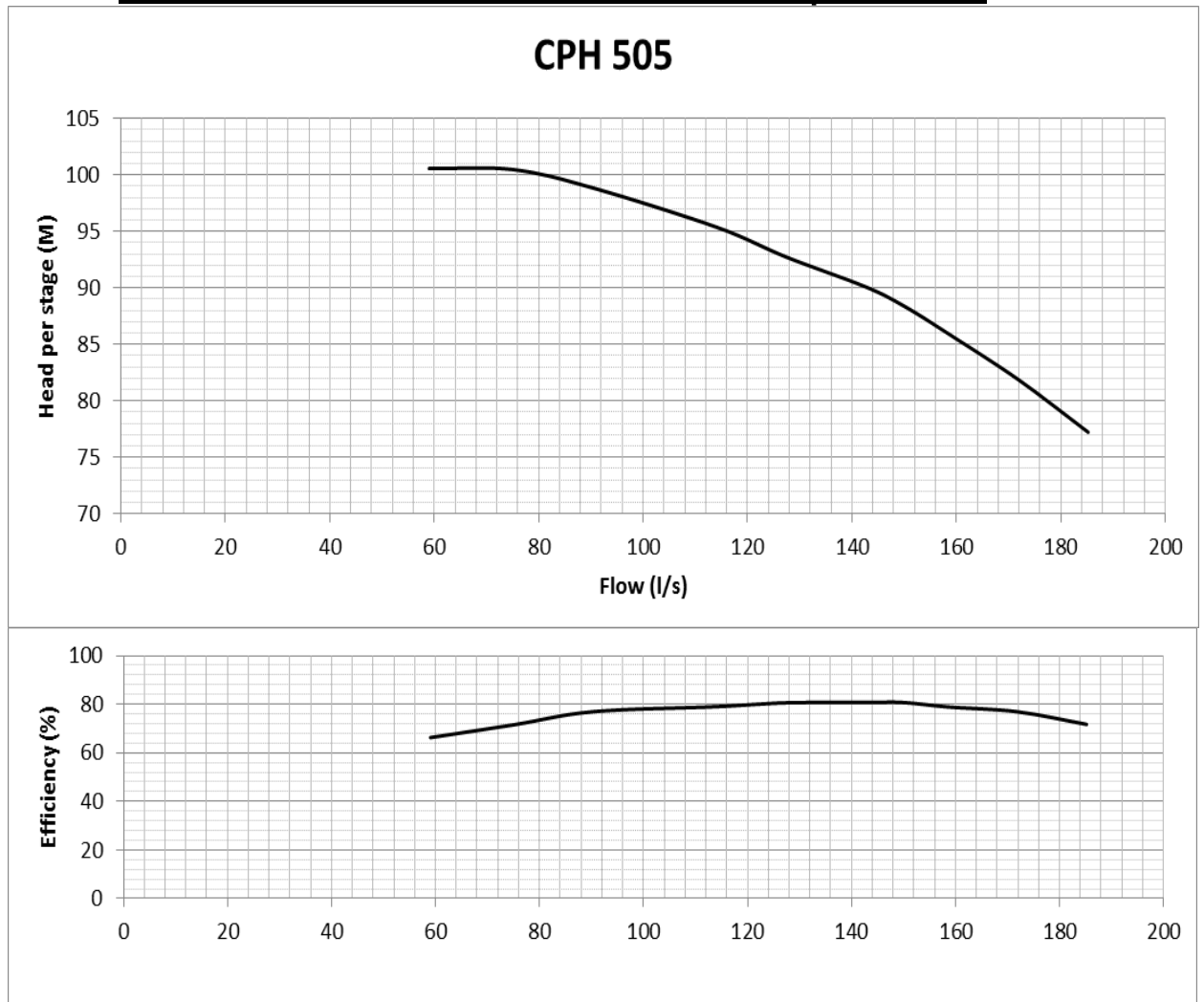
- For Balance disc assembly protection, fully adjustable wear limit switch with motor cut out sensors / pump motor trip function
- Bearing temperature probes
- Horizontal plane drive end bearing vibration monitoring probe

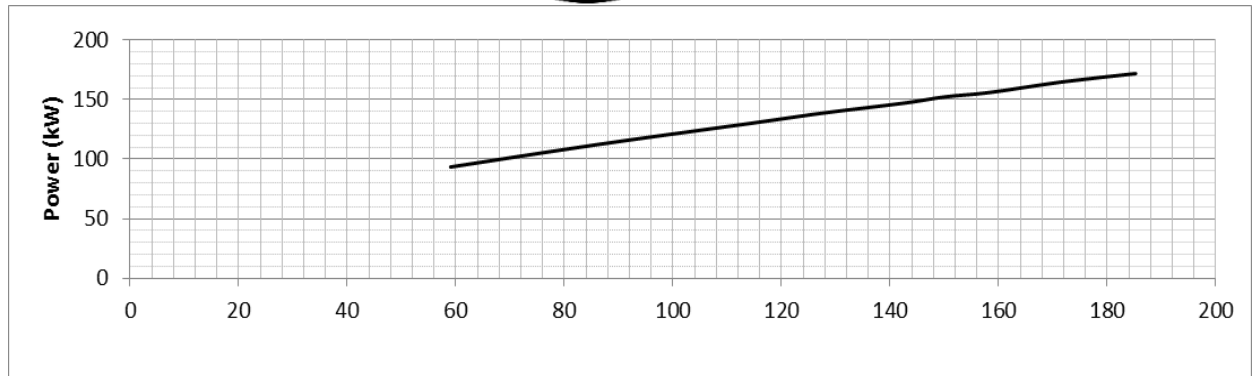
Services:

On site commissioning and monitoring both locally and globally by certified pump technicians with all on site monitoring equipment and telemetry



## Cemo CPH 505 – Tested Pump Curve





|                          |          |                        |         |
|--------------------------|----------|------------------------|---------|
| <b>Pump speed</b>        | 1490 rpm | <b>Inlet diameter</b>  | 225mm   |
| <b>Impeller Diameter</b> | Ø505mm   | <b>Outlet diameter</b> | 200mm   |
| <b>NPSH (min)</b>        | 11m      | <b>Q @ BEP</b>         | 138 l/s |